AI ENGINEERING JOURNAL

Team name: NgaChainAILayer2
AI Tools used: Claude Code (Claude Opus 5.5) as pair engineer and as a separate strict reviewer; its add-ons for UI critique, landing-page design and browser testing. Playwright for the automated browser test and the screenshots we reviewed. GitHub CLI for the repository. Microsoft Word to export this journal.
Language: English

1. AI Strategy

Briefly describe how your team used AI during the project.

(Max. 150 words).

We used AI as a fast pair engineer and as a strict reviewer, never as the decision maker. Every change had to trace back to our Round 1 proposal: AI read the Round 2 brief and our deck and proposed a narrow MVP (one Swedish DC, two rules, one end-to-end journey); we chose the stack, the rule of no AI inside the product, the visual direction and the Vietnamese option. AI wrote code together with tests, and we only accepted what we could check: 46 automated tests, including a real-browser run of the journey, and screenshots of every screen. Before submitting we ran a separate AI review told to be strict. It found real gaps, such as an on-time KPI that ignored overdue orders and journey times that did not match the activity log; we fixed them and added tests. Documents are generated from the running app.

2. AI Usage log

Document the most significant AI interactions during your project. Maximum 5 tasks

Stage	Purpose of using AI	Original time estimate	Actual time with AI	Human Validation	Outcome
Scoping	Read the Round 2 brief and our Round 1 deck; propose a narrow MVP scope and stack	1 day	3 h	We chose the stack and 'no AI in the product'; scope checked against Round 1 slides 4-8	One DC, two rules, one journey; FastAPI + SQLite

Stage	Purpose of using AI	Original time estimate	Actual time with AI	Human Validation	Outcome
Build	Implement source adapter, rules, escalation, workflow, KPIs and demo harness, with tests	4 days	1 day	A test per rule; demo run end to end against Round 1 slide 4	Working MVP; stale-feed bug (9 false alarms) fixed at the root
UI and language	Generate UI options, build the timetable view, add English and Vietnamese	3 days	1 day	We picked the direction from real screens; Vietnamese checked by the team	Timetable UI; English and Vietnamese
Documents	Build the system diagram and the user journey from the running app	1.5 days	4 h	Rejected overlapping auto-layouts; every arrow checked against code	Two 2-page PDFs, times from the app
Review and hardening	Strict AI review of the whole submission, then fixes and a refactor	2 days	5 h	We triaged each finding; re-ran 46 tests and the demo	Overdue counts as late; cost log; scorecard; refactor

3. Key AI Decisions

Describe **2–3 examples** where AI had a meaningful impact on your project.

Task	AI suggestion	What your team did
Data freshness guard	First version: pause alerts only when the warehouse feed is older than 30 minutes.	Screenshots of an outage showed 9 false alarms at minute 30. We asked for a root-cause rule: a missing event only counts if the feed was read after the milestone. Tests cover both cases.
Honest outcome measurement	The strict review showed Outcomes at 100% on time at 14:30 while SE-1077 was past its 14:00 promise and not delivered.	We changed the rule: an order past its promise without a delivery is overdue and counts as late. Percentages from fewer than 10 cases are no longer marked on or below target.
AI inside the product	Offered an optional AI case summary and an ML late-risk score.	We kept the product rules-only, as Round 1 argued, and made every closed exception a labelled example (GET /api/outcomes) for the risk model planned after Gate 2.

4. Reflection

Answer each question in 2-4 sentences

1. Where did AI help your team the most?

   In the build-and-check loop. AI turned our Round 1 rules into working code with tests within hours, and its browser screenshots let us review every state in both languages without clicking through each one. It was also useful as a second pair of eyes: the strict review found mismatches between our documents and the app that we had stopped noticing.

2. What AI output did you reject or modify, and why?
   We rejected automatically laid-out C4 diagrams because labels overlapped and lines crossed boxes; they are now laid out by hand from the code structure. We rejected the first stale-data rule after it produced false alarms. We declined an AI summary feature, and we changed AI's first KPI definitions, which made the demo look better than an honest measure would (for example, counting false alarms as orders saved).

3. What engineering or business decisions required human judgment?
   Keeping AI out of the product was a business decision: Round 1 asked Company A to fund explainable rules first. Decision rights (only the shift lead may expedite), the promise that can never be edited, showing EUR 25 only as a Round 1 assumption, and what the demo is allowed to claim were our calls. We also decided which review findings to fix and which to leave.

4. Are there any obstacles when using AI?
   AI sometimes declared work finished while the rendered screen still had overlaps or a misleading number, so screenshot review became mandatory. It writes plausible but unverified claims, so we checked every sentence in the documents against the code and the running app. Long sessions lose context, so we kept a short written product brief that every session read first.